

WALVIS BAY, Namibia

WMO: 680980

Lat: 22.980S

Lon: 14.645E

Elev: 91

StdP: 100.24

Time Zone: 2.00 (E02)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.8	6.8	-9.1	1.7	21.1	-6.0	2.3	22.8	12.4	25.1	10.3	21.6	2.4	60	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	10.2	35.0	18.1	32.9	16.8	30.8	16.9	21.3	29.4	20.3	27.6	19.5	26.2	6.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.8	24.6	17.9	13.0	23.3	17.0	12.3	22.0	62.3	29.6	58.7	27.8	55.9	26.3	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	8.7	7.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.3	20.6	20.7	21.0	20.3	18.7	17.5	17.4	14.5	15.5	16.7	17.8	19.2
	DBStd	3.72	1.62	1.88	2.99	4.08	4.04	4.99	4.85	2.67	1.88	1.76	1.72	1.50
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	538	1	1	3	18	43	78	82	128	89	57	28	8
	CDD10.0	3033	327	301	341	311	268	225	229	139	164	207	235	286
	CDD18.3	530	71	68	86	78	53	54	53	9	4	6	13	36
	CDH23.3	4144	207	264	624	777	581	577	611	141	68	69	110	115
	CDH26.7	1653	25	47	257	357	264	275	285	66	22	18	27	9
(14) Wind	WSAvg	3.3	3.6	3.4	3.1	3.0	2.9	3.4	3.5	3.1	3.2	3.4	3.4	3.4
(15) Precipitation	PrecAvg	34	5	6	9	3	0	1	0	0	1	1	2	3
	PrecMax	80	32	41	47	25	5	3	8	3	11	5	12	15
	PrecMin	7	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	24	8	9	10	7	1	1	2	1	2	1	3	4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.1	31.6	37.6	37.1	36.1	36.0	34.2	33.2	29.3	31.4	30.1	28.1
		MCWB	20.0	21.0	21.4	19.7	17.6	16.3	15.8	15.1	14.4	18.1	18.9	18.1
	2%	DB	27.0	28.0	33.2	33.8	32.9	33.1	32.8	27.6	24.2	25.5	25.8	26.1
		MCWB	19.1	19.6	20.9	19.1	16.6	15.0	15.3	13.4	15.0	16.2	16.5	17.7
	5%	DB	25.6	26.2	29.2	30.6	29.2	30.0	30.2	22.9	21.9	23.1	24.0	24.8
		MCWB	18.6	19.0	19.4	18.8	16.3	14.6	14.8	12.5	14.2	15.3	16.1	17.5
	10%	DB	24.2	25.0	26.4	27.7	26.0	26.2	26.9	20.1	20.1	21.6	22.7	23.5
		MCWB	18.2	18.6	19.0	17.9	15.3	13.4	13.9	13.1	14.2	15.0	15.7	17.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.5	21.5	23.7	22.2	20.3	17.3	18.7	15.9	16.6	18.9	18.9	19.4
		MCDB	27.0	29.2	31.5	29.8	28.5	31.1	26.2	26.6	24.5	29.3	25.0	25.6
	2%	WB	19.7	20.3	21.6	20.8	18.0	16.1	17.3	14.6	15.6	17.1	17.3	18.4
		MCDB	25.2	26.5	29.8	30.1	27.9	28.5	25.3	23.0	22.4	24.1	23.6	24.6
	5%	WB	19.1	19.6	20.4	19.6	16.9	15.2	16.0	14.0	15.0	16.0	16.6	17.8
		MCDB	24.4	25.2	28.1	28.7	27.2	26.1	25.0	20.5	20.6	22.1	22.5	23.6
	10%	WB	18.5	18.9	19.1	18.2	16.1	14.5	14.9	13.3	14.2	15.2	15.9	17.3
		MCDB	23.2	24.0	25.8	26.4	24.3	23.7	25.2	19.4	19.2	20.6	21.7	22.8
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.7	10.2	11.9	13.2	13.6	13.6	12.5	11.0	10.4	10.5	9.0
		MCDBR	9.6	10.3	14.0	15.3	17.0	16.4	17.6	17.9	15.0	13.5	12.9	10.9
	5% WB	MCWBR	3.8	4.0	5.0	5.6	6.4	7.1	7.8	8.7	7.1	6.0	5.7	4.4
		MCWBR	8.3	9.4	12.1	13.3	14.6	14.8	15.2	13.5	11.9	11.4	11.1	9.4
(40) Clear-Sky Solar Irradiance	taub		0.436	0.443	0.416	0.371	0.340	0.328	0.348	0.393	0.456	0.454	0.400	0.408
		taud	2.254	2.257	2.314	2.402	2.424	2.414	2.362	2.233	2.025	2.082	2.284	2.279
	Ebn at Noon		911	891	888	887	876	866	857	850	836	871	938	938
		Edh at Noon	148	145	131	110	99	95	104	129	172	170	143	145
(44) All-Sky Solar Radiation	RadAvg		7.56	7.09	6.34	5.38	4.50	4.11	4.29	4.88	5.71	6.73	7.68	7.86
	RadStd		0.31	0.30	0.23	0.24	0.17	0.14	0.13	0.22	0.25	0.21	0.35	0.28

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page